

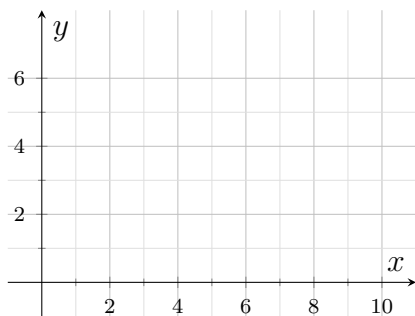
# Perimeter and Area of Coordinate Polygons

*Finding a polygon perimeter from coordinates, finding a polygon area from coordinates, and using slope or midpoint to justify a shape before measuring it*

**The three tools, and when each one earns its place.**

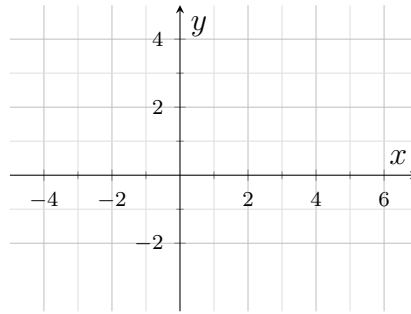
- **Distance** gives one side length:  $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ . A perimeter is the sum of the side lengths, so find the ones you cannot read off the grid and add.
- **Shoelace** gives the area of any polygon straight from its vertices in order:  $A = \frac{1}{2} |\sum (x_i y_{i+1} - x_{i+1} y_i)|$ . Splitting the polygon into right triangles and rectangles gives the same number.
- **Slope and midpoint** do not measure anything. They justify a claim about the shape — perpendicular sides, bisected diagonals — which is often what makes a shortcut legal.
- Plot the polygon before computing. Every problem gives you a grid for it, and a sketch catches a mis-copied coordinate in seconds.
- Leave exact radicals unless the problem names a unit to round to, and label areas in square units.

**1.** A triangular sail is stitched along three straight seams. On the design grid its corners are  $A(2, 1)$ ,  $B(10, 1)$  and  $C(10, 7)$ , and one grid unit represents one metre. Plot the sail, find the length of the slanted seam  $AC$ , then find the total length of seaming around the sail.



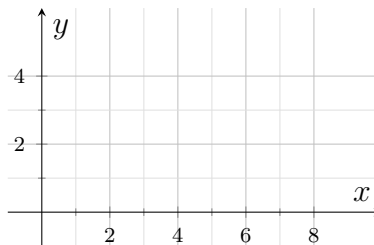
**Answer:** \_\_\_\_\_ m

- 2.** A rectangular concrete slab has corners at  $(-3, -2)$ ,  $(5, -2)$ ,  $(5, 3)$  and  $(-3, 3)$  on a site plan where one grid unit represents one metre. Plot the slab and find its area.



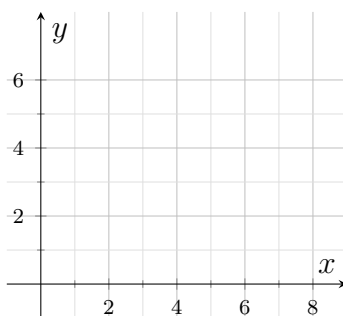
**Answer:** \_\_\_\_\_  $\text{m}^2$

- 3.** A parallelogram-shaped floor tile has vertices  $P(0, 0)$ ,  $Q(6, 0)$ ,  $R(9, 4)$  and  $S(3, 4)$ , measured in grid units. Plot the tile, find the length of side  $QR$ , then find the perimeter of the tile.



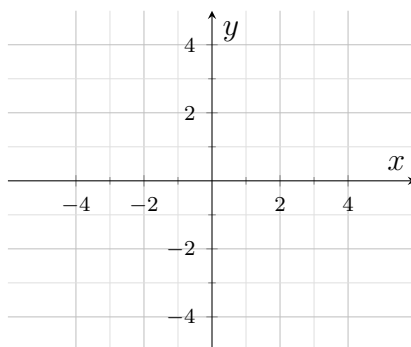
**Answer:** \_\_\_\_\_ units

4. A triangular flowerbed has corners at  $(0, 0)$ ,  $(8, 2)$  and  $(3, 7)$  in grid units. Plot the flowerbed and find its area.



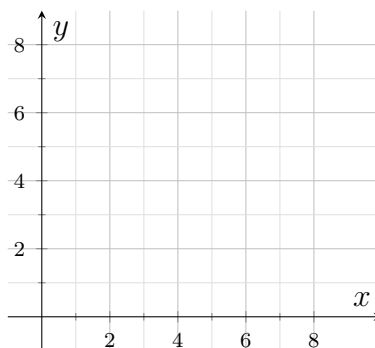
**Answer:** \_\_\_\_\_ square units

5. A kite-shaped banner has corners  $A(-4, 0)$ ,  $B(0, 3)$ ,  $C(4, 0)$  and  $D(0, -3)$ , in grid units. Plot it, then find the midpoint of diagonal  $AC$  and the midpoint of diagonal  $BD$ . Find the length of side  $AB$ . All four sides of this banner are the same length — use that to find its perimeter.



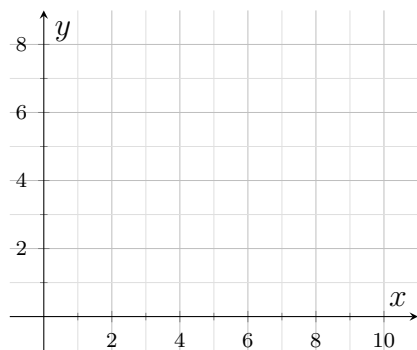
**Answer:** \_\_\_\_\_ units

- 6.** A triangular park has corners  $L(1, 2)$ ,  $M(5, 0)$  and  $N(9, 8)$ , in grid units. Plot the park, then find the slope of  $LM$  and the slope of  $MN$ . Say in one sentence what those two slopes tell you about the corner at  $M$ , and use it to find the area of the park.



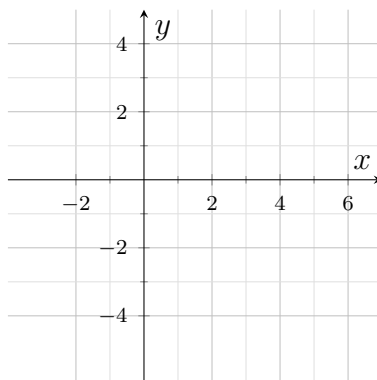
**Answer:** \_\_\_\_\_ square units

- 7.** A building lot has corners at  $(0, 0)$ ,  $(10, 0)$ ,  $(10, 6)$  and  $(2, 8)$  on a survey grid where one grid unit represents one metre. Plot the lot and find its area.



**Answer:** \_\_\_\_\_ m<sup>2</sup>

8. A designer claims that the quadrilateral with vertices  $W(-2, 1)$ ,  $X(3, 3)$ ,  $Y(5, -2)$  and  $Z(0, -4)$ , in grid units, is a square. Plot it and find the length of side  $WX$  in exact radical form, then find the area of the quadrilateral. Congruent sides on their own would make the figure only a rhombus, so finish with a short written justification, using slopes, that decides whether the designer's claim is correct.



**Answer:** \_\_\_\_\_ square units